

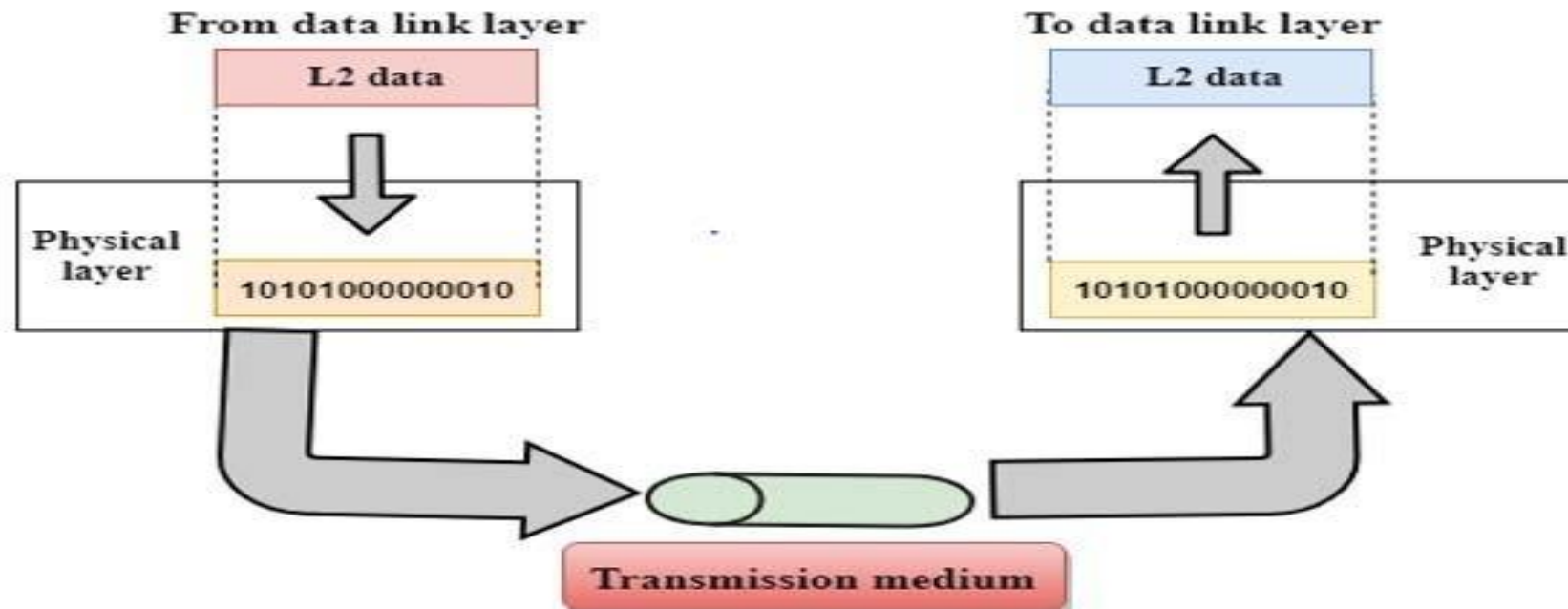
Unit 2

Physical Layer

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2.1 Functions of Physical Layer

- The physical layer consists of all the functions required to transmit a bitstream over a physical medium.
- Physical layer devices or layer 1 devices are Hub, Repeater, Modem, Cables.



Functions of Physical Layer

1. Representation of bits:

The physical layer data consists of a stream of bits (a sequence of 0s and 1s) without any interpretation.

To be transmitted, bits must be encoded into the signals - electrical or optical, or waves – radio or microwaves, etc.

✓ The physical layer defines the type of encoding (how 0's and 1's are changed in signals).

✓ Physical layer encodes data in the form of electrical signals, optical signals, or radio waves.

✓ The physical layer is responsible for delivering those signals from a cable, an optical fiber, or a Wi-Fi router.

Functions of Physical Layer

2. Data Rate (Transmission Rate):

- ✓ The number of bits sent each second is also defined by the physical layer.
- ✓ In other words, the physical layer defines the duration of a bit also.

3. Synchronization of bits:

- ✓ It is necessary to have synchronization between the sender and receiver at the bit level.

Functions of Physical Layer

4. Line configuration:

✓ Physical layer is concerned with the connection of devices to the channel or medium, whether point-to-point or multi-point.

5. Physical topology:

✓ The physical topology determines how devices are connected to create a network.

✓ Devices can be using a mesh topology, a star topology, a ring topology or a bus topology

Functions of Physical Layer

6. Transmission mode:

- ✓ The mechanism of transferring data or information between two linked devices connected over a network is referred to as transmission mode.
- ✓ They are simplex, half-duplex, or full-duplex.

7. Physical characteristics of media (Transmission medium):

- ✓ It represents the types of transmission medium i.e; guided medium or unguided medium.

2.2 Data and Signals

The term analog data refers to information that is continuous, and digital data refers to information that has discrete states.

To be transmitted, data must be transformed into electromagnetic signals.

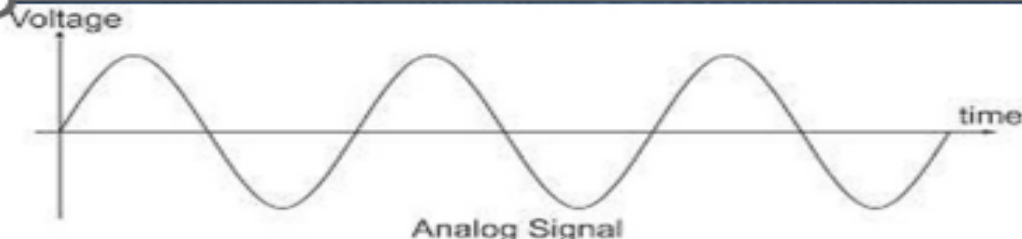
A signal is an electromagnetic or electrical current that is used for carrying data from one system or network to another.

A message encoded using electric current is called an electronic signal

Data and Signals

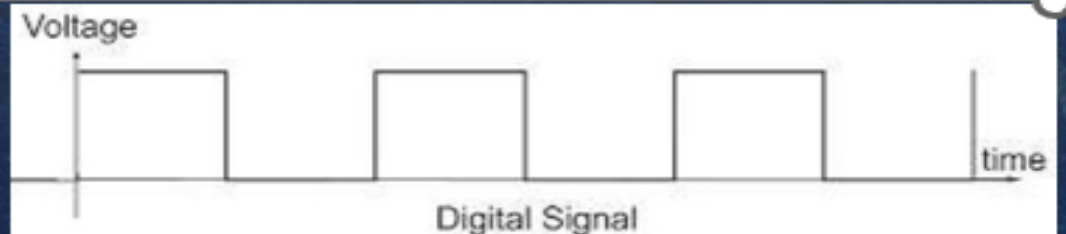
Analog Signal

- An analog signal is a continuous signal that represents physical measurements.
- It is denoted by sine waves.
- It uses a continuous range of values that help to represent information.
- The analog signal bandwidth is low.
- Temperature sensors, FM radio signals, Light sensor, are examples of Analog signals.



Digital Signal

- Digital signals are time separated signals which are generated using digital modulation.
- It is denoted by square waves.
- Digital signal uses discrete 0 and 1 to represent information.
- The digital signal bandwidth is high.
- Computers, CDs, DVDs are some examples of Digital signal.

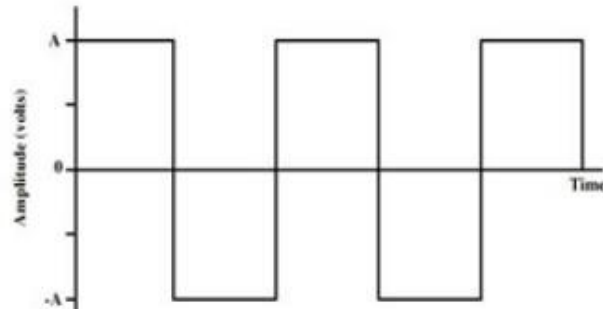
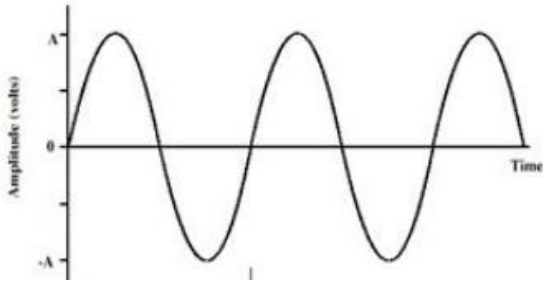


Data and Signals

Analog and digital signals can be in two forms:

1. Periodic Signal

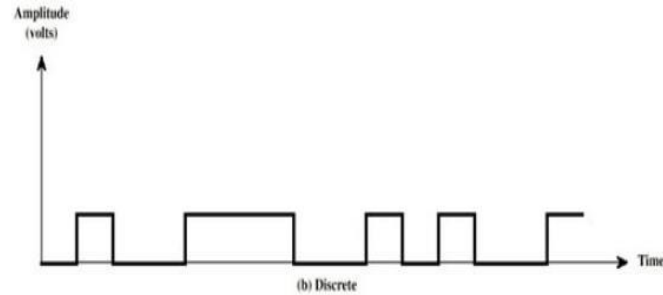
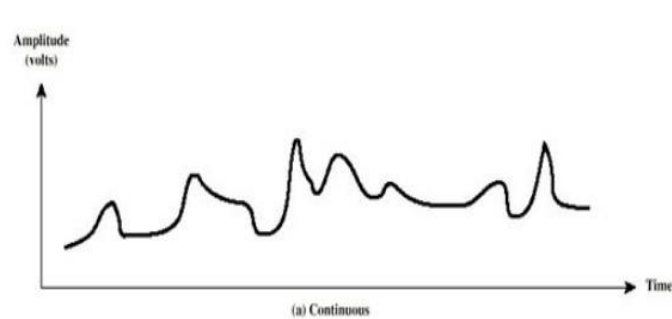
A periodic signal completes a pattern within a measurable time frame, called a period, and repeats that pattern over subsequent identical periods.



Data and Signals

2. Non-periodic signal:

A non-periodic signal is a signal that does not repeat over time.



Data and Signals

2. Transmission impairments

The signals that are transmitted at the beginning of the medium are not the same as the signals that are received at the end of the medium, i.e., what is sent is not what is received, due to various transmission impairments.

These impairments tend to degrade the quality of analog and digital signals.

Consequences

- ✓ For a digital signal, there may occur bit errors.
- ✓ For analog signals, these impairments degrade the quality of the signals.

Data and Signals

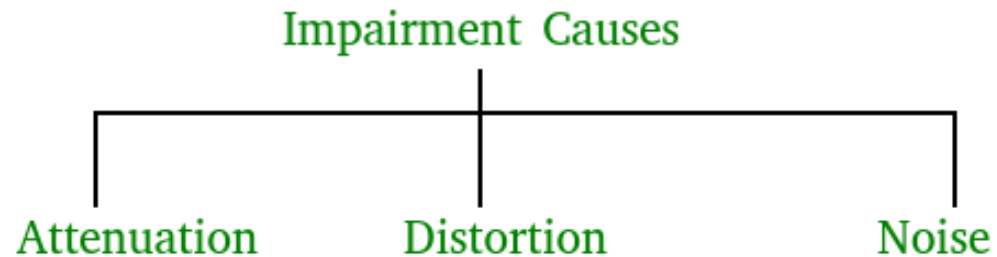
Causes of impairment

There are three main causes of impairment:

a) Attenuation

b) Distortion

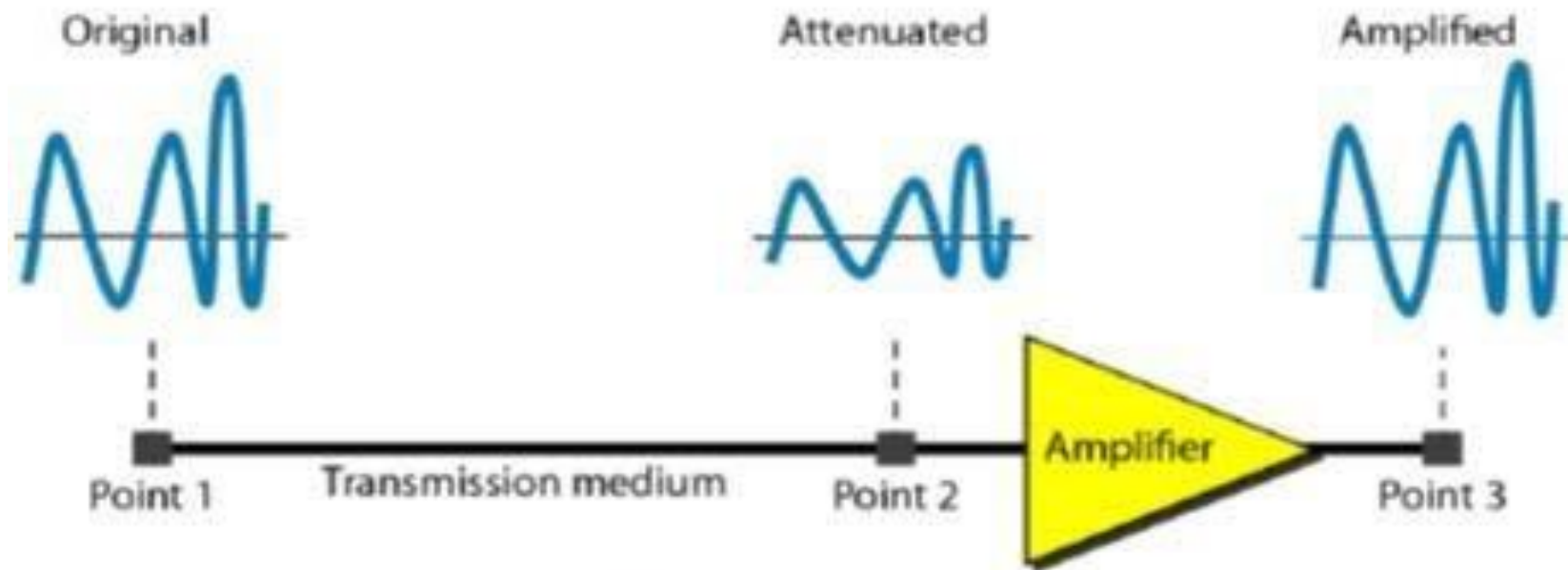
c) Noise



a. Attenuation

- ✓ Attenuation means loss of energy.
 - ✓ The strength of the signal decreases with increasing distance, which causes a loss of energy in overcoming the resistance of the medium. This is also known as an attenuated signal.
 - ✓ Amplifiers are used to amplify the attenuated signal which gives the original signal back, and compensate for this loss.
- Attenuation is measured in decibels(dB). It measures the relative strengths of two signals or one signal at two different points.
- ✓ $\text{Attenuation(dB)} = 10\log(P2/P1)$
 - ✓ P1 is the power at the sending end, and P2 is the power at Receiving end.

a. Attenuation



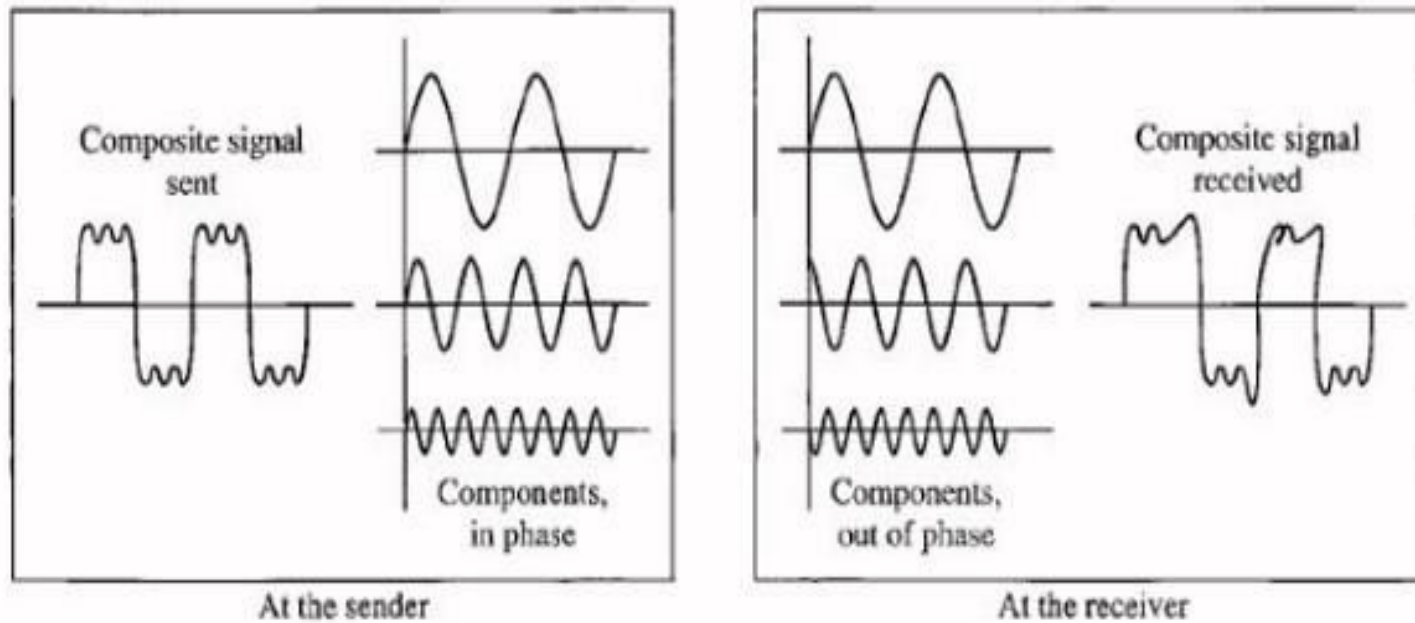
Decibel: Measure the **relative power** (attenuation)

$$\text{dB} = 10 \log_{10} P_2 / P_1$$

b. Distortion

- ✓ Distortion means signal changes its form or shape.
- ✓ Distortion can occur in a composite signal made up of different frequencies.
- ✓ Each component of frequency has its propagation speed traveling through a medium and therefore, different components have different delay in arriving at the final destination.
- ✓ Equalization (removal of distortion) techniques can be used to smooth out the distortion.

c. Distortion



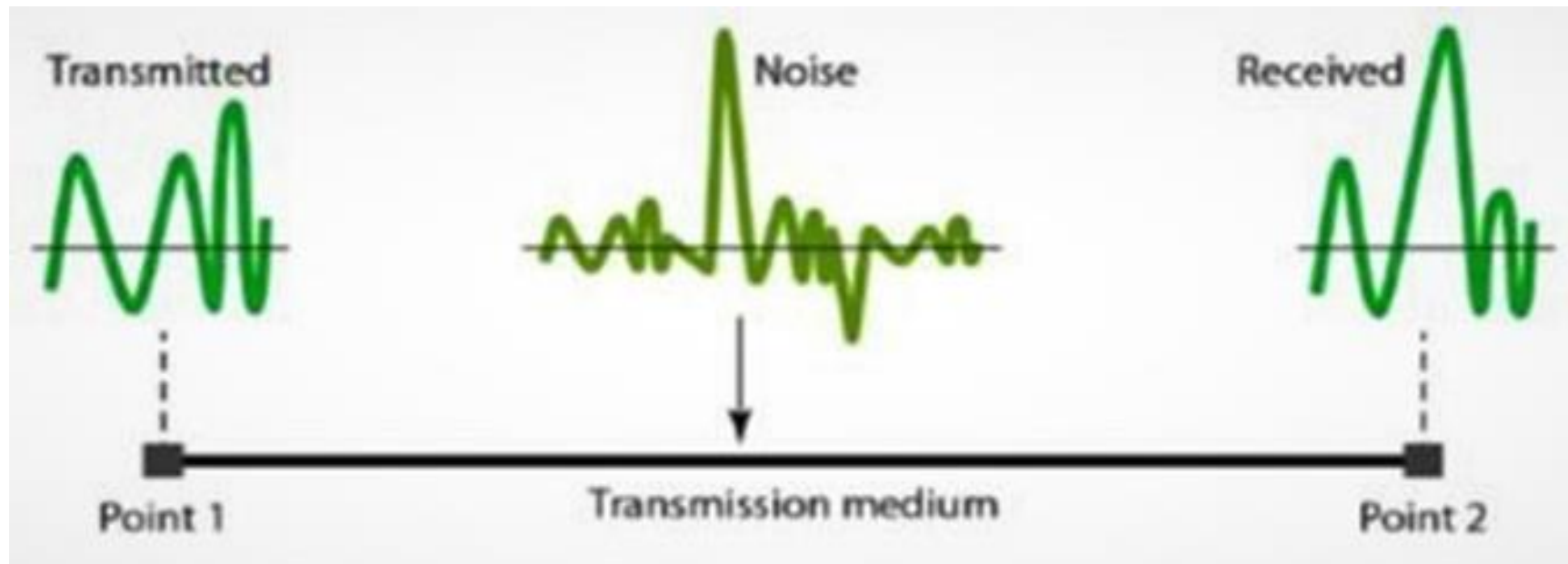
Distortion

Noise

- ✓ The random or unwanted signal that mixes up with the original signal is called noise.
- ✓ There are several types of noise such as induced noise, crosstalk noise, thermal noise and impulse noise which may corrupt the signal. Proper use of shielded cables help minimizing noise.
- ✓ Induced noise comes from sources such as motors and appliances. These devices act as sending antenna and transmission medium act as receiving antenna.
- ✓ Crosstalk noise is when one wire affects the other wire.
- ✓ Thermal noise is movement of electrons in wire which creates an extra signal.
- ✓ Impulse noise is a signal with high energy that comes from lightning or power lines
- ✓ The signal-to-noise ratio is defined as

$$\text{SNR} = \text{AVG SIGNAL POWER} / \text{AVG NOISE POWER}$$

Noise



3. Data Rate Limits

- ✓ Data rate refers to the speed of data transfer through a channel.
- ✓ It is generally computed in bits per second (bps).
- ✓ One of the main objectives of data communications is to increase the data rate.
- ✓ Three factors determine the data rate of a channel:
 - Bandwidth of the channel(directly proportional)
 - Number of levels of signals that are used(directly prop)
 - Noise present in the channel(inversely proportional)
- ✓ Data rate can be calculated using two theoretical formulae:
 - Nyquist Bit Rate – for a noiseless channel
 - Shannon's Capacity – for a noisy channel

a) Noiseless Channel: Nyquist Bit Rate

$$\text{Bit Rate} = 2 * \text{Bandwidth} * \log_2(L)$$

Where Bandwidth is the bandwidth of the channel,

L is the number of signal levels used to represent data, and

Bit Rate is the bit rate in bits per second.

✓ Bandwidth is a fixed quantity, so it cannot be changed. Hence, the data rate is directly proportional to the number of signal levels.

Note: Increasing the level of a signal may reduce the reliability of the system.

b) Noisy Channel: Shannon Capacity

In reality, we cannot have a noiseless channel; the channel is always noisy.

Shannon capacity is used to determine the theoretical highest data rate for a noisy channel as:

$$\text{Capacity} = \text{Bandwidth} * \log_2(1 + \text{SNR})$$

Where Bandwidth is the bandwidth of the channel,

SNR is the **signal-to-noise ratio**, and

Capacity is the capacity of the channel in bits per second.

Bandwidth is a fixed quantity, so it cannot be changed.

Hence, the channel capacity is directly proportional to the power of the signal, as **SNR = (Power of signal) / (power of noise)**.

Example

Q1. A telephone line normally has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communication. The SNR is usually 3162. What will be the capacity for this channel?

$$C = 3000 * \log_2(1 + \text{SNR}) = 3000 * 11.62 = 34881.23 \text{ bps}$$

Q2. The SNR is often given in decibels. Assume that SNR(dB) is 36 and the channel bandwidth is 2 MHz. Calculate the theoretical channel capacity.

$$\text{SNR(dB)} = 10 * \log_{10}(\text{SNR})$$

$$\log_{10}(\text{SNR}) = \text{SNR(dB)}/10$$

$$\text{SNR} = 10^{(\text{SNR(dB)}/10)} = 10^{(36/10)}$$

$$\text{SNR} = 10^{3.6} = 3981$$

$$\text{Hence, } C = 2 * 10^6 * \log_2(3982) = 24 \text{ MHz}$$

Example

Q3. Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel the capacity C is calculated as

$$C = B \log_2 (1 + \text{SNR}) \quad C = B \log$$

$$_2(1 + 0) \quad C = B \log_2(1) \quad (2^x=1)$$

$$C = B \times 0 \quad C = 0$$

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot receive any data through this channel.

4. Performance

- ✓ Performance of a network refers to the **measure of service quality** of a network as perceived by the user.
- ✓ The characteristics that measure the performance of a network are :
 - a) Bandwidth
 - b) Throughput
 - c) Latency (Delay)
 - d) Bandwidth – Delay Product
 - e) Jitter

4. Performance

a. Bandwidth

- ✓ Bandwidth is a measure of how much information a network can transfer.
- ✓ Bandwidth describes the maximum data transfer rate of a network.
- ✓ Bandwidth of analog and digital signals is calculated in separate ways:

Bandwidth of an Analog Signal (in Hz)

If the signal has a minimum frequency of $F1 = 1000\text{Hz}$ and maximum frequency of $F2 = 5000\text{Hz}$. Hence, the bandwidth is given by $F2 - F1 = 5000 - 1000 = 4000\text{ Hz}$

Bandwidth of a Digital Signal (in bps)

For example, we can say the bandwidth of a fast Ethernet network is a maximum of 100 Mbps, which means that the network can send 100 Mbps of data.

4. Performance

b. Throughput

The throughput is a measure of how fast we can actually send data through a network.

In other words,

The bandwidth is a potential measurement of a link.

The throughput is an actual measurement of how fast we can send data.

Input: A network with a bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute where each frame carries an average of 10,000 bits. What will the throughput of this network be?

Output: We can calculate the throughput as

$$\text{Throughput} = (12,000 \times 10,000) / 60 = 2 \text{ Mbps}$$

The throughput is nearly equal to one-fifth of the bandwidth in this case.

4. Performance

c) Latency (Delay) The total time taken for a complete message to arrive at the destination, starting with the time when the first bit of the message is sent out from the source and ending with the time when the last bit of the message is delivered at the destination.

Latency = Transmission Time + Propagation Time + Queuing Time + Processing Delay

- i. Transmission delay – time taken to push the packet's bits onto the link
- ii. Propagation delay – time for a signal to propagate through the media
- iii. Queuing delay – time the packet spends in routing queues
- iv. Processing delay – time taken by a router to process the packet header (Header – contains information about the packet, including origin, destination, length and packet number.)

4. Performance

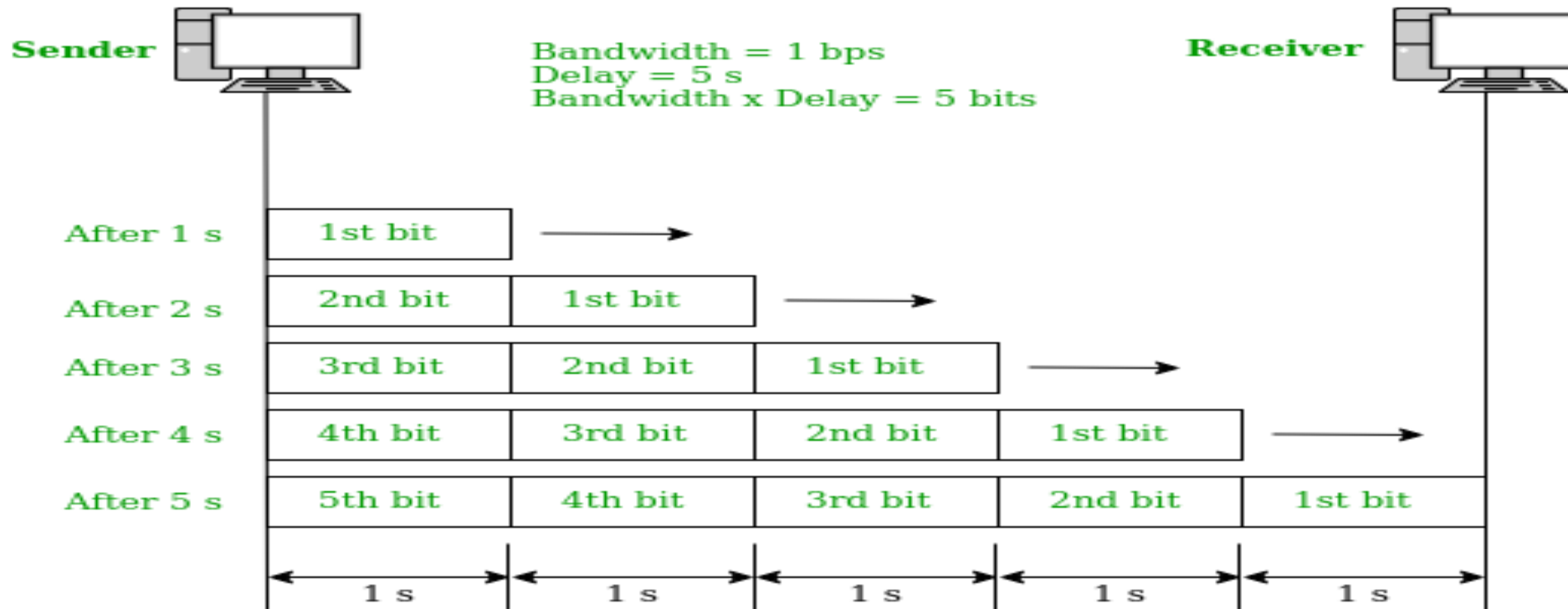
d) Bandwidth Delay Product

Bandwidth delay product is a measurement of how many bits can fill up a network link i.e; capacity of the link.

✓ It gives the maximum amount of data that can be transmitted by the sender at a given time (unitary method, if user can send 1 bit in 1 second-1 bps-bandwidth then he/she can send 5bits in 5sec, 6bits in 6sec etc).

✓ Assume a link is of bandwidth 1 bps and the delay of the link is 5s. Let us find the bandwidth-delay product in this case. We can say that this product $1 \times 5 = 5\text{bits}$ is the maximum number of bits that can fill the link. There can be close to 5 bits at any time on the link.

4. Performance



4. Performance

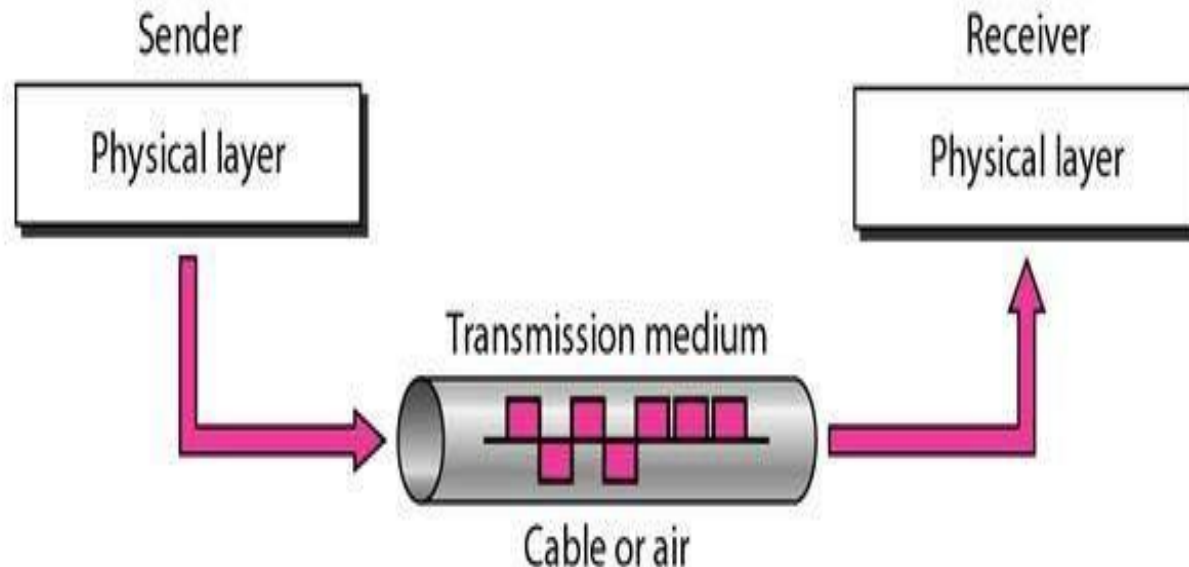
e) Jitter

- ✓ Jitter is defined as a variation in the delay of received packets.
- ✓ Jitter is considered as a problem when different packets of data face different delays in a network and the data at the receiver application is time-sensitive, i.e. audio or video data.
- ✓ Jitter is measured in milliseconds(ms).

2.3 Data Transmission Media: Guided Media, Unguided Media and Satellites

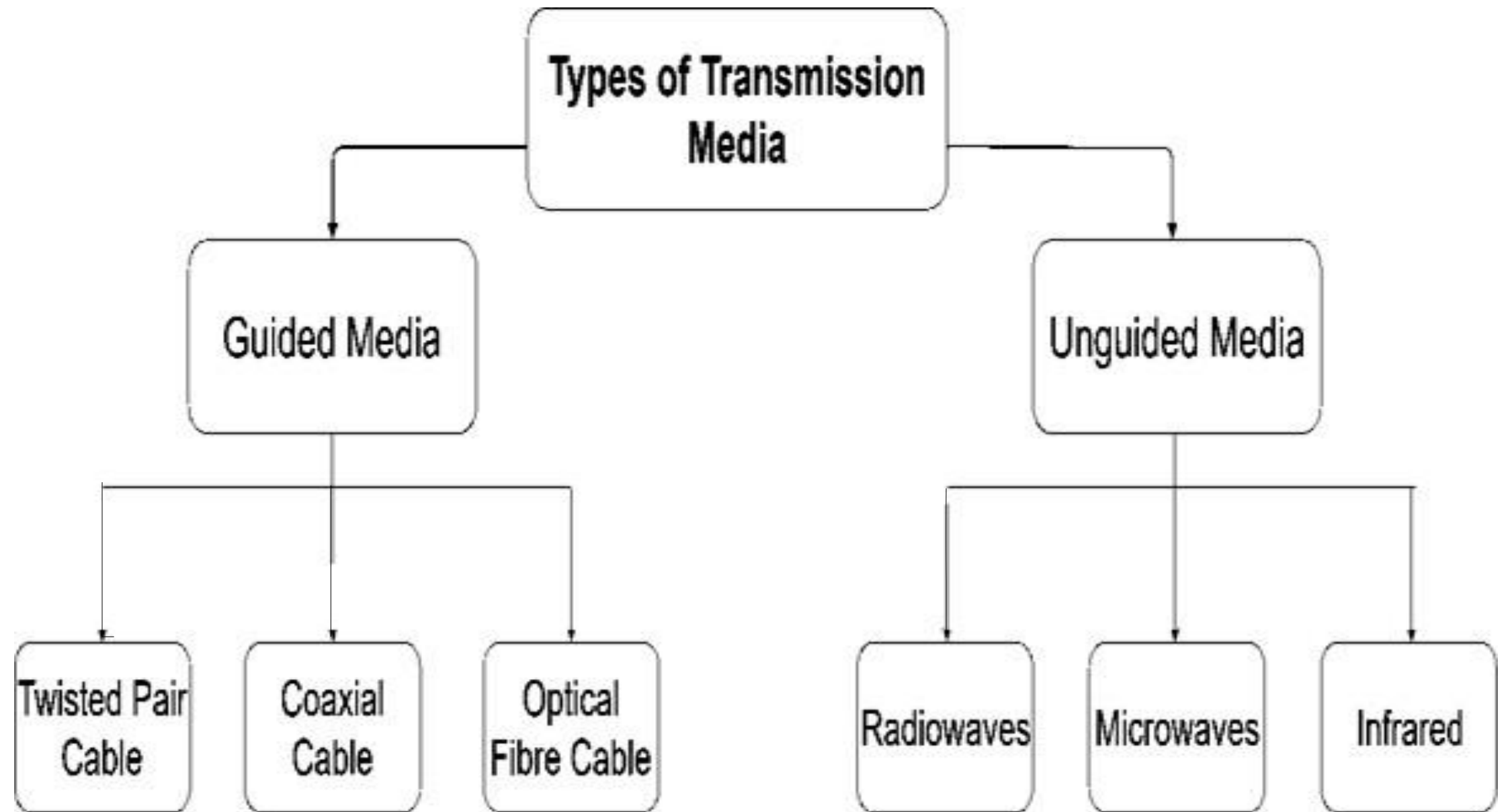
Transmission media are actually located below the physical layer and are directly controlled by the physical layer.

The following figure shows the position of transmission media in relation to the physical layer.



Data Transmission Media

- ✓ A transmission medium can be broadly defined as anything that can carry information from a source to a destination.
- ✓ In telecommunications, transmission media can be divided into two broad categories: guided and unguided.
 1. Guided media include twisted-pair cable, coaxial cable, and fiber-optic cable.
 2. Unguided medium is free space



Data Transmission Media

a) Twisted-pair cable

✓ Twisted pair is a physical media made up of a pair of cables twisted with each other.

transmission media.

✓ Installation of the twisted pair cable is easy, and it is a lightweight cable.

Twisted Pair is of two types:

i. Unshielded Twisted Pair (UTP):

✓ This type of cable has the ability to block interference and does not depend on a physical shield for this purpose.

✓ It is used for telephonic applications.

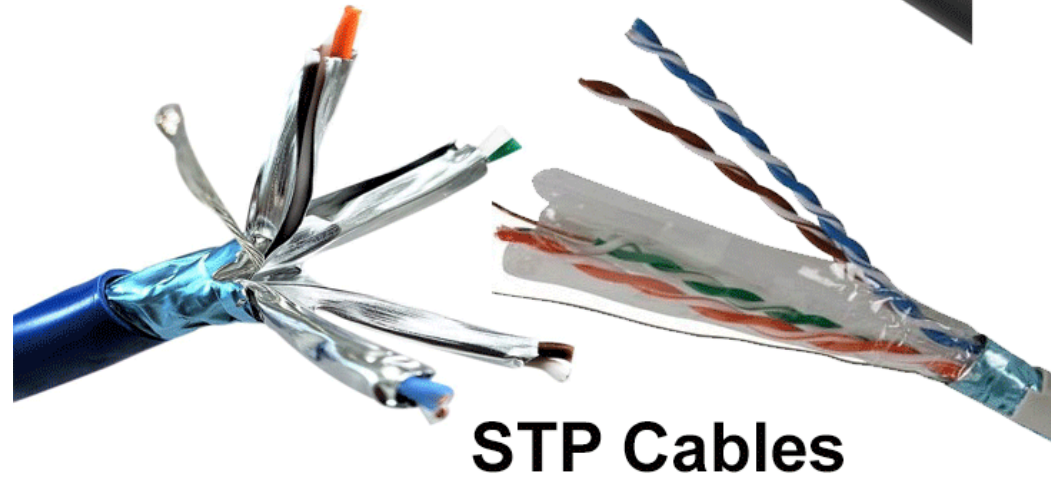
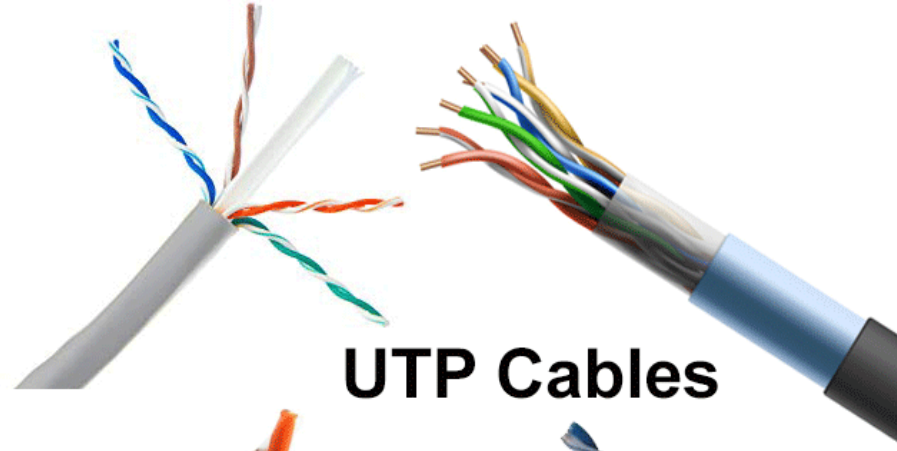
Advantages of UTP

- ✓ Least expensive
- ✓ Easy to install
- ✓ High speed capacity

Disadvantages of UTP

- ✓ Susceptible to external interference
- ✓ Lower capacity and performance in comparison to STP
- ✓ Short distance transmission due to attenuation

UTP and STP



ii. Shielded Twisted Pair (STP):

- ✓ This type of cable consists of a special jacket to block external interference.
- ✓ It is used in fast-data-rate Ethernet.

Advantages:

- ✓ Better performance at a higher data rate in comparison to UTP
- ✓ Eliminates crosstalk
- ✓ Comparatively faster

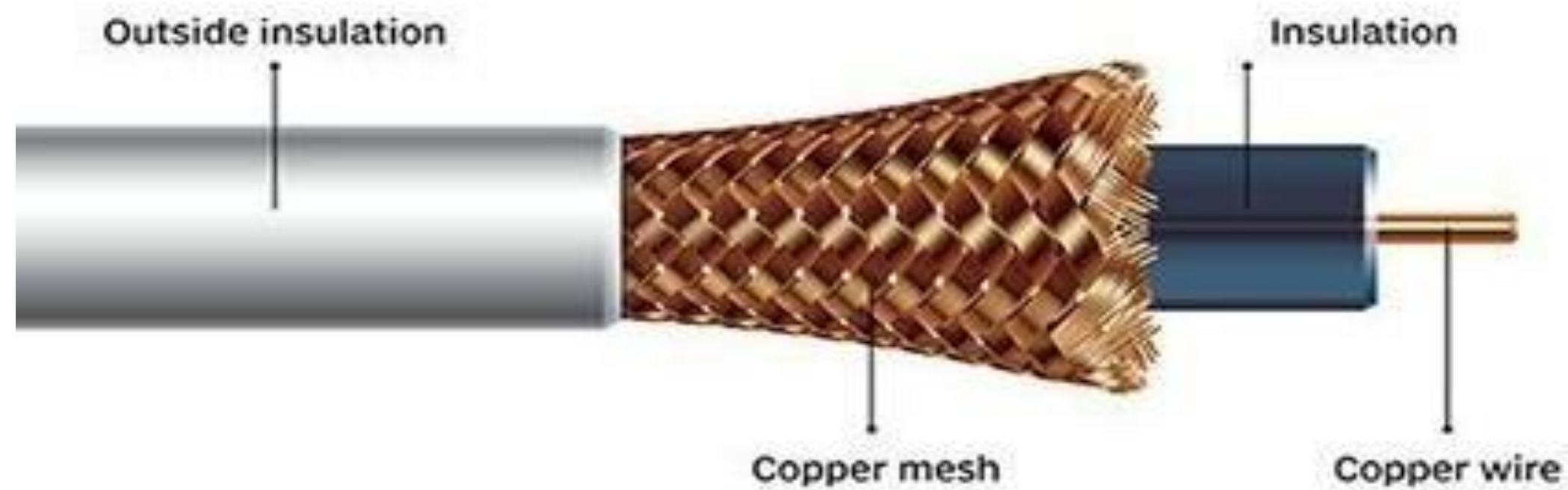
Disadvantages:

- ✓ Comparatively difficult to install and manufacture
- ✓ More expensive
- ✓ Bulky

b) Coaxial Cable

- ✓ Coaxial cable is very commonly used transmission media, for example, TV wire is usually a coaxial cable.
- ✓ The inner conductor of the coaxial cable is made up of copper, and the outer conductor is made up of copper mesh. The middle core is made up of non-conductive cover that separates the inner conductor from the outer conductor.
- ✓ The inner core is responsible for the data transferring whereas the copper mesh prevents from the EMI(Electromagnetic interference).

Coaxial cable



Coaxial cable transmits information in two modes:

- ✓ Baseband mode: It is defined as the process of transmitting a single signal at high speed.
- ✓ Broadband mode: It is defined as the process of transmitting multiple signals simultaneously.

Advantages:

- ✓ High Bandwidth
- ✓ Better noise immunity
- ✓ Easy to install and expand
- ✓ Inexpensive

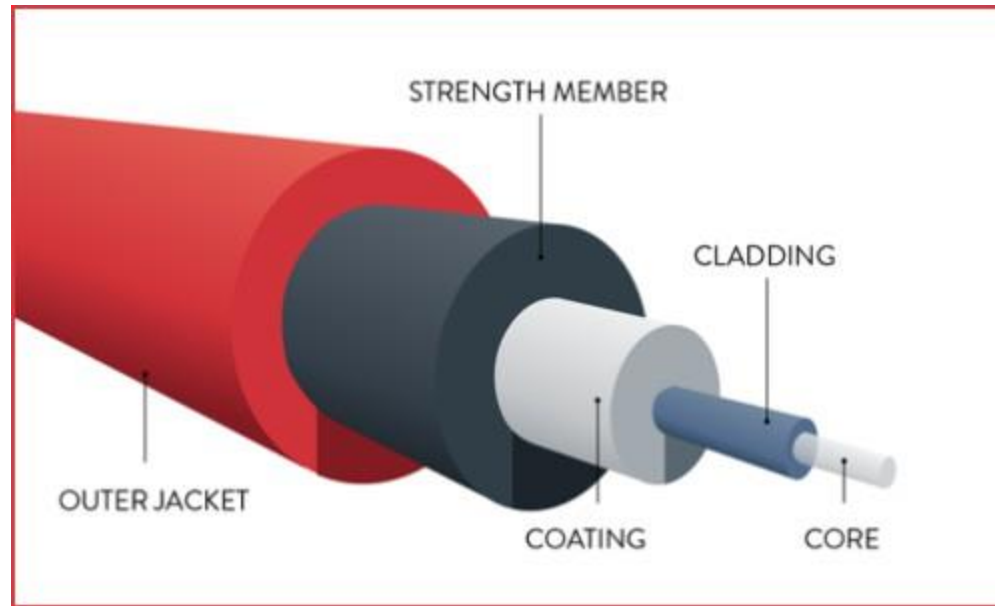
Disadvantages:

- ✓ Single cable failure can disrupt the entire network

c) Optical Fibre Cable

- ✓ It uses the concept of reflection of light through a core made up of glass or plastic.
- ✓ The core is surrounded by a less dense glass or plastic covering called the cladding.
- ✓ It is used for transmission of large volumes of data.

Optical Fibre Cable



Optical Fibre Cable

Advantages:

- ✓ Increased capacity and bandwidth
- ✓ Light weight
- ✓ Less signal attenuation
- ✓ Immunity to electromagnetic interference

Disadvantages:

- ✓ Difficult to install and maintain
- ✓ High cost
- ✓ Fragile(easily broken or damaged)

2. Unguided Media

- ✓ It is also referred to as Wireless or Unbounded transmission media.
- ✓ No physical medium is required for the transmission of electromagnetic signals.

Features:

- ✓ The signal is broadcasted through air
- ✓ Less Secure
- ✓ Used for larger distances

There are 3 types of waves transmitted through unguided media:

(i) Radio Waves

- ✓ Electromagnetic waves ranging in frequencies between 3KHz and 1GHz are called radio waves.
- ✓ These are easy to generate and can penetrate through buildings.
- ✓ AM and FM radios, bluetooth and Wi-Fi use radiowaves for transmission.

Advantages:

- ✓ Wires are not needed as they travel through air, thus a cheaper form of communication.

Disadvantages:

- ✓ Travel in a straight line(reflected by barrier), so repeater stations may be needed.

(ii) Micro Waves

- ✓ Electromagnetic waves having frequencies between 1GHz and 300GHz are called micro waves.
- ✓ It is a line of sight transmission i.e. waves travel in a direct path from the source to the receiver.
- ✓ These are majorly used for mobile phone communication.

Advantages:

- ✓ Wires are not needed as they travel through air, thus a cheaper form of communication.

Disadvantages:

- ✓ Travel in a straight line, so repeater stations may be needed.

(iii) Infrared Waves

- ✓ Electromagnetic waves having frequencies between 300GHz and 400THz are called infrared waves.
- ✓ Infrared waves are used for very short distance communication.
- ✓ They cannot penetrate through obstacles.
- ✓ It is used in TV remotes.

Advantages:

- ✓ Used in scientific experimentation to identify the heat of an object.

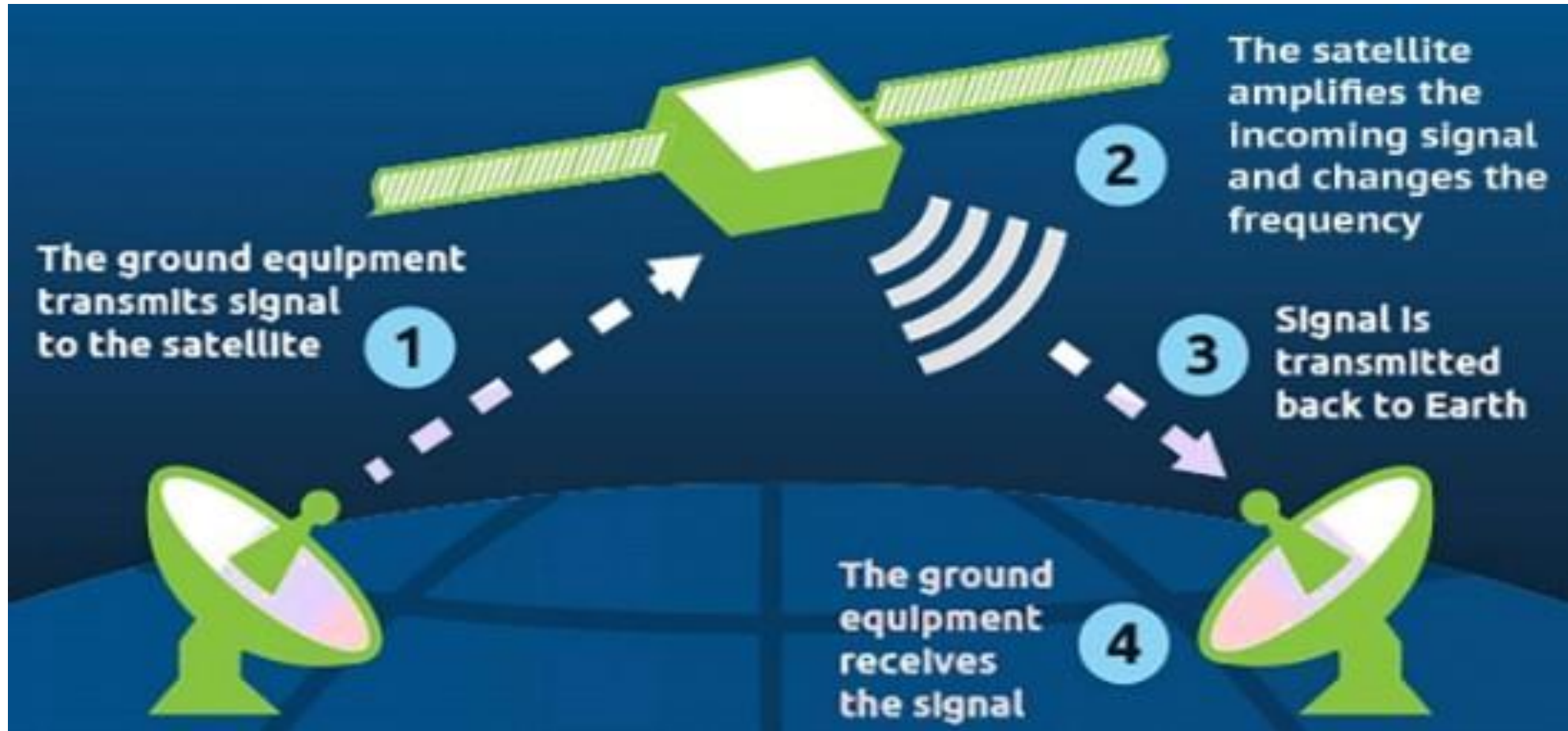
Disadvantages:

- ✓ Can cause minor burns if exposed to skin for long periods of time.

Satellites

- ✓ A communications satellite is an artificial satellite that relays and amplifies radio telecommunication signals via a transponder(wireless device that picks up and automatically responds to an incoming signal).
- ✓ It creates a communication channel between a source transmitter and a receiver at different locations on Earth.
- ✓ Communications satellites are used for television, telephone, radio, internet, and military applications.
- ✓ The purpose of communications satellites is to relay the signal around the curve of the Earth allowing communication between widely separated geographical points.

Satellites



Satellites

Advantages:

- ✓ Access to remote areas
- ✓ Coverage of large geographical areas
- ✓ High bandwidth

Disadvantages:

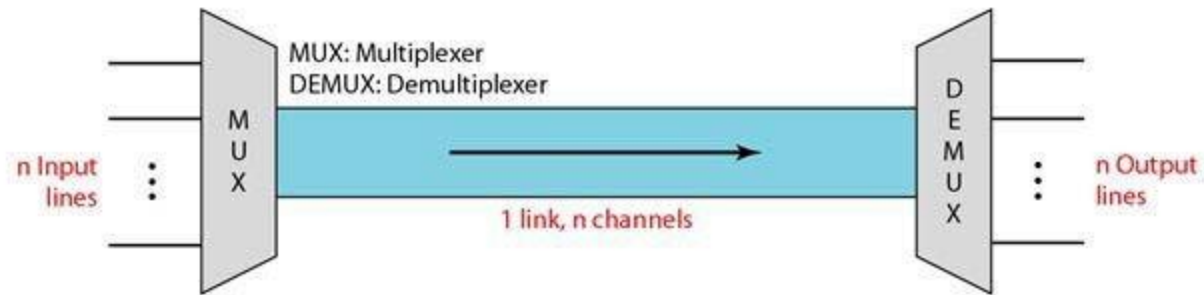
- ✓ High initial cost
- ✓ Propagation delay with GEO systems
- ✓ Environmental interference problems

2.4 Bandwidth Utilization: Multiplexing and Spreading

Multiplexing

- ✓ Multiplexing is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link.
- ✓ Whenever the bandwidth of a medium linking two devices is greater than the bandwidth needs of the devices, the link can be shared.
- ✓ In a multiplexed system, n lines share the bandwidth of one link.
- ✓ The following figure shows the basic format of a multiplexed system.

Multiplexing



Multiplexing

- ✓ The lines on the left direct their transmission streams to a multiplexer (MUX), which combines them into a single stream (many-to-one).
- ✓ At the receiving end, that stream is fed into a demultiplexer (DEMUX), which separates the stream back into its component transmissions (one-to-many) and directs them to their corresponding lines.

Advantages of Multiplexing:

- ✓ More than one signal can be sent over a single medium.
- ✓ The bandwidth of medium can be utilized effectively.

The three basic multiplexing techniques are

1. Frequency-division multiplexing (FDM)
2. Wavelength-division multiplexing (WDM)
3. Time-division multiplexing (TDM)

Frequency-division multiplexing (FDM)

Frequency Division Multiplexing (FDM) is a communication technique where multiple signals are transmitted simultaneously over a single communication channel by dividing the available bandwidth into separate frequency ranges.

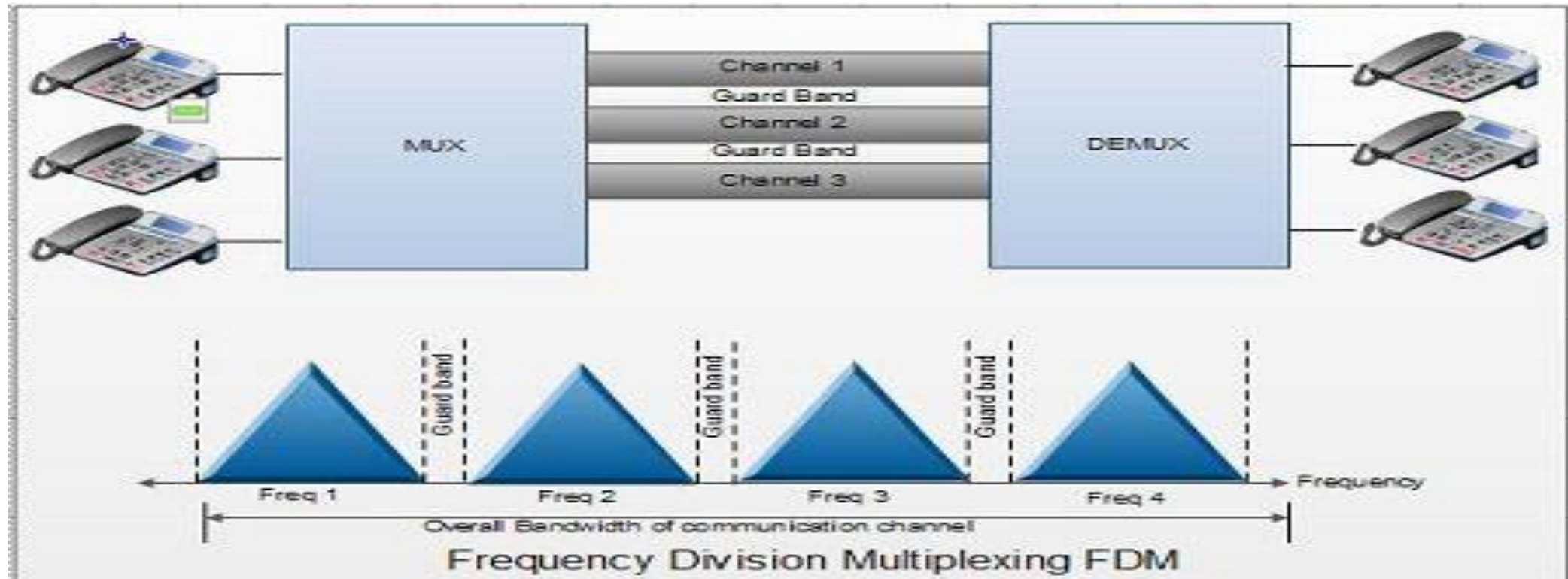
Frequency-division multiplexing (FDM)

- The total bandwidth of a channel is split into smaller non-overlapping frequency bands.
- Each signal (like voice, video, or data) is assigned its own unique frequency band.
- All signals are sent at the same time, but because they use different frequencies, they don't interfere with each other.
- At the receiver end, filters separate the signals back into their original form.

Frequency-division multiplexing (FDM)

- Many radio stations broadcast at the same time.
- Each station uses a different frequency (e.g., 91.1 MHz, 98.3 MHz).
- Your radio tunes into one frequency to listen to a specific station.

Frequency-division multiplexing (FDM)



Wavelength-division multiplexing (WDM)

Wavelength Division Multiplexing (WDM) is a technique used in **fiber-optic communication** where multiple data signals are transmitted.

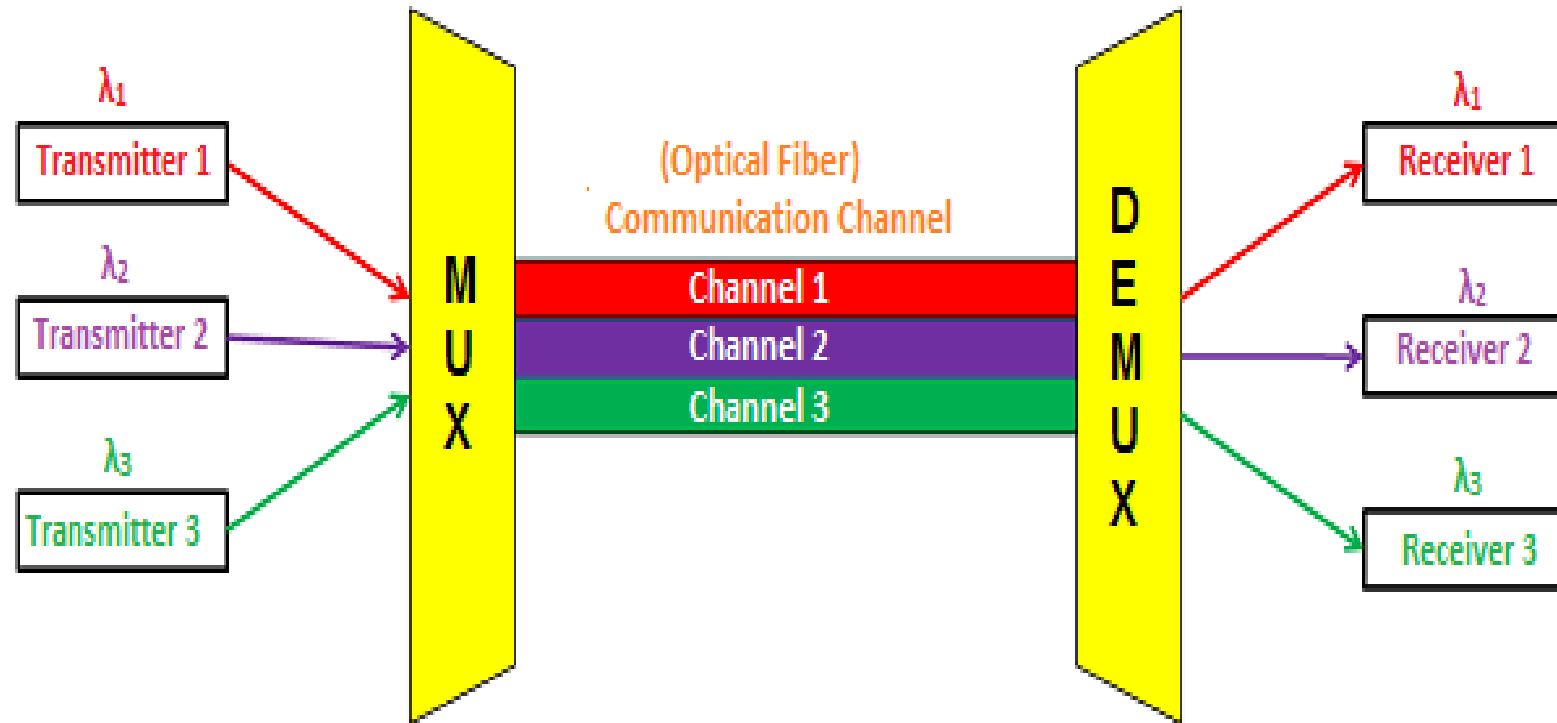
Each data signal is converted into a light signal with a specific wavelength.

Multiple light beams (each with a different wavelength) are combined into one fiber.

At the receiving end, the combined signal is split back into individual wavelengths.

Each wavelength carries its own independent data stream.

Wavelength-division multiplexing (WDM)



Wavelength Division Multiplexing

Time-division multiplexing (TDM)

Time Division Multiplexing (TDM) is a technique where multiple signals share a single communication channel by taking turns transmitting in **different time slots**.

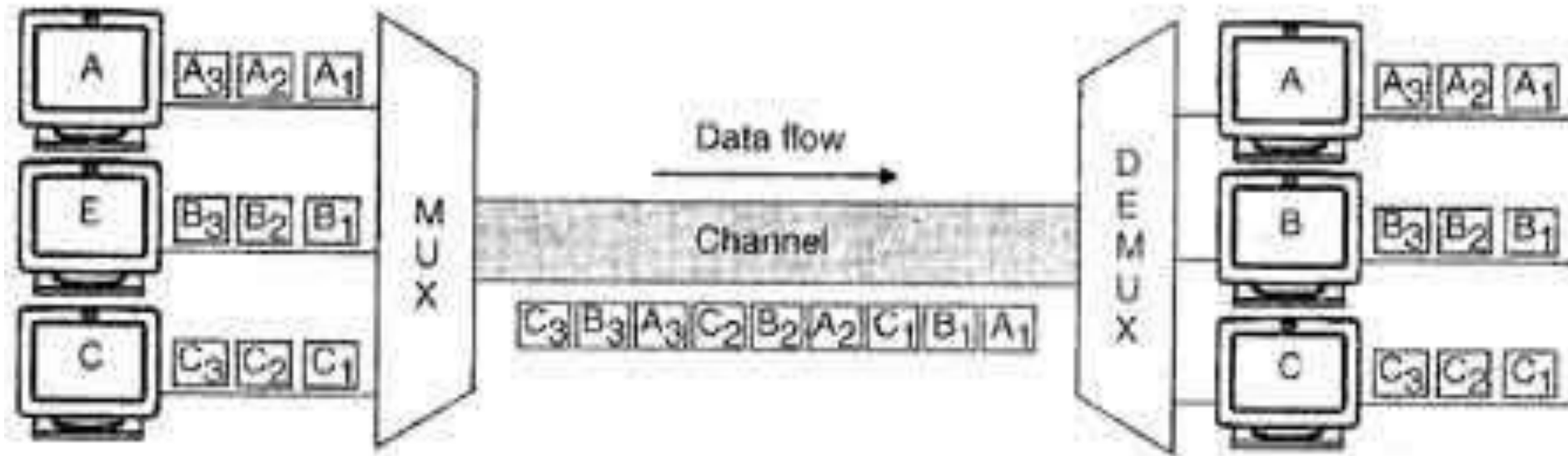
The entire time on a channel is divided into small, repeating time slots.

Each signal (user/device) is assigned a specific time slot.

Signals are sent one after another very quickly, so it appears simultaneous.

At the receiver, the signals are separated based on their time slots.

Time-division multiplexing (TDM)



Time Division Multiplexing (TDM)

Spread Spectrum

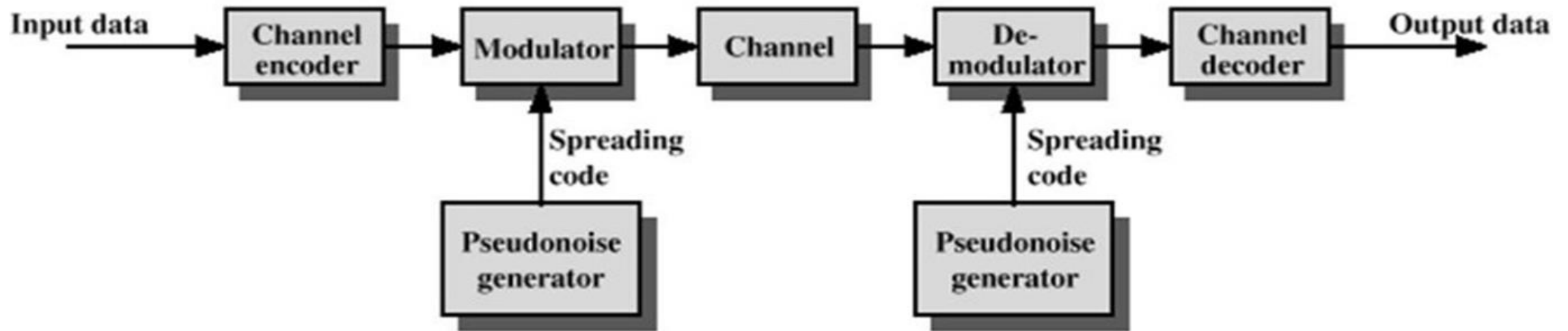
Spread Spectrum is a communication technique in which a signal is spread over a **wide range of frequencies**, much wider than the minimum bandwidth required for transmission.

Instead of putting all your message in one “channel,” you **spread it across many channels**

It was initially developed for military and intelligence requirements.

Spread spectrum is designed to be used in wireless applications.

The use of spread spectrum makes jamming and interception more difficult.



Input is fed into a channel encoder that produces a signal.

This signal is modulated using a sequence of digits known as a spreading code generated by a pseudo noise generator, which increases the bandwidth of the signal to be transmitted.

On the receiving end, the same digit sequence is used to demodulate the spread spectrum signal, which is then fed into a channel decoder to recover the data.

1. Frequency Hopped Spread Spectrum (FHSS)

In this technique, the users are made to change the frequencies of usage from one to another in a specified time interval.

For example, a frequency was allotted to sender 1 for a particular period of time and after a while sender 1 hops to the other frequency and sender 2 uses the first frequency which was previously used by sender 1. This is called as frequency reuse.

The frequencies of the data are hopped from one to another in order to provide a secure transmission.

The amount of time spent on each frequency hop is called as Dwell time.

2. Direct Sequence Spread Spectrum (DSSS)

- ✓ In this technique, each and every bit of the user data is multiplied by a secret code called chipping code.
- ✓ This chipping code is the spreading code which is multiplied with the original message and transmitted.
- ✓ The receiver uses the same code to retrieve the original message.

Advantages of Spread Spectrum:

- ✓ Cross-talk elimination
- ✓ Better output with data integrity
- ✓ Better security
- ✓ Hard to detect
- ✓ Difficult to jam the signals

2.5 Switching: Circuit switching, Message switching & Packet switching

- ✓ Switching is the process to forward packets coming in from one port to a port leading towards the destination.
- ✓ When data comes on a port it is called **ingress**, and when data leaves a port or goes out it is called **egress**.

Switching can be divided into two major categories:

- ✓ Connectionless: No previous handshaking is required.
- ✓ Connection Oriented: Before switching data to be forwarded to destination, there is a need to pre-establish circuit along the path between both endpoints. Data is then forwarded on that circuit.

Switching Methods

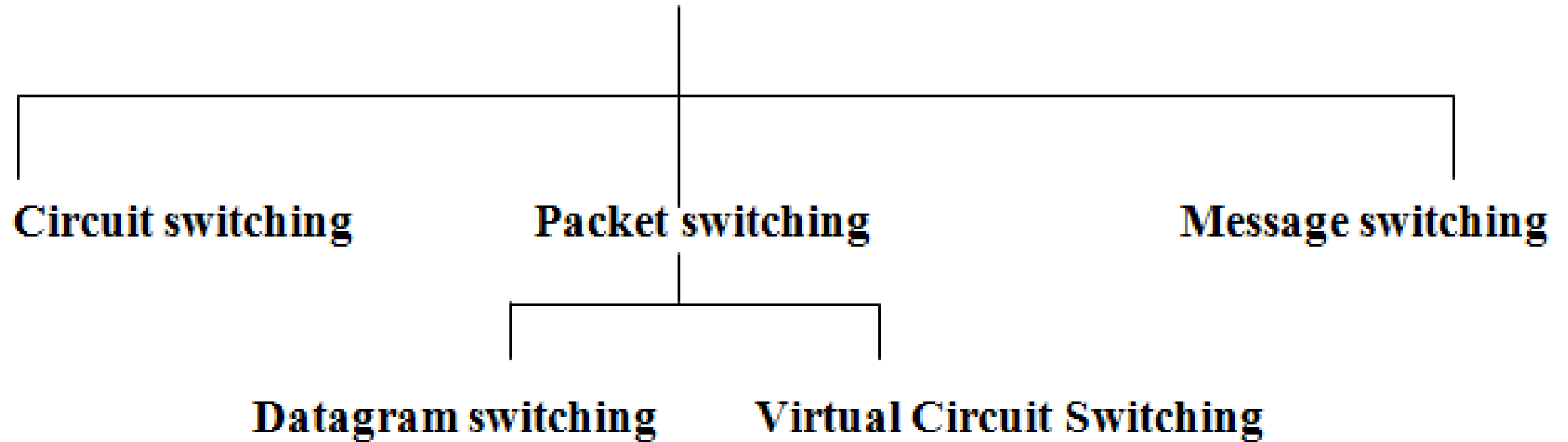
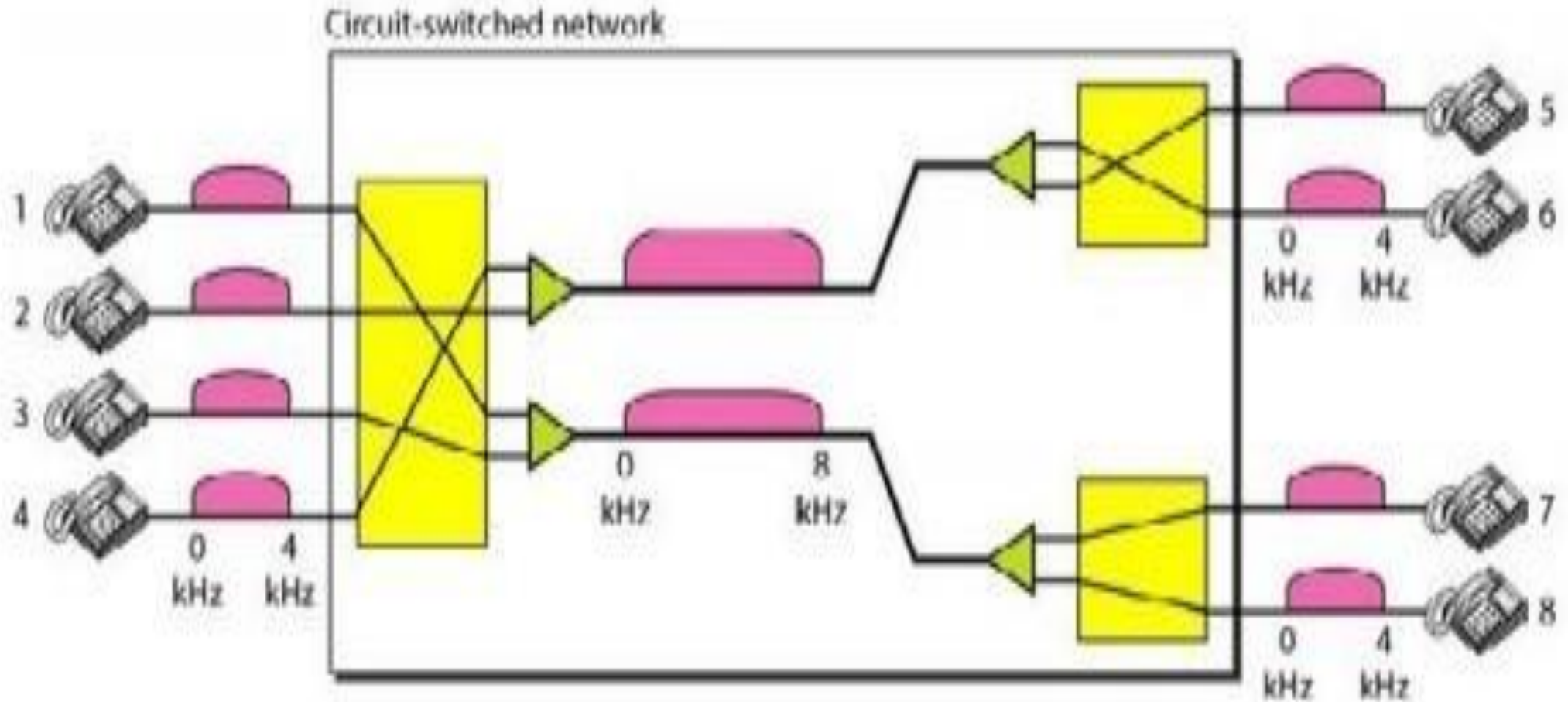


Fig- Types of switching methods

Circuit Switching

- ✓ When two nodes communicate with each other over a dedicated communication path, it is called circuit switching.
- ✓ There is a need of pre-specified route from which data will travel and no other data is permitted.
- ✓ In circuit switching, to transfer the data, circuit must be established so that the data transfer can take place.
- ✓ Applications which use circuit switching will have to go through three phases:
 - Circuit establishment
 - Data transfer
 - Circuit disconnect

Circuit Switching



Circuit Switching

✓ Telephone is the best suitable example of circuit switching.

Advantages:

✓ It is suitable for long continuous transmission.

✓ The dedicated path ensures a steady communication.

Disadvantages:

✓ Bandwidth requirement is high even in cases of low data volume.

Message switching

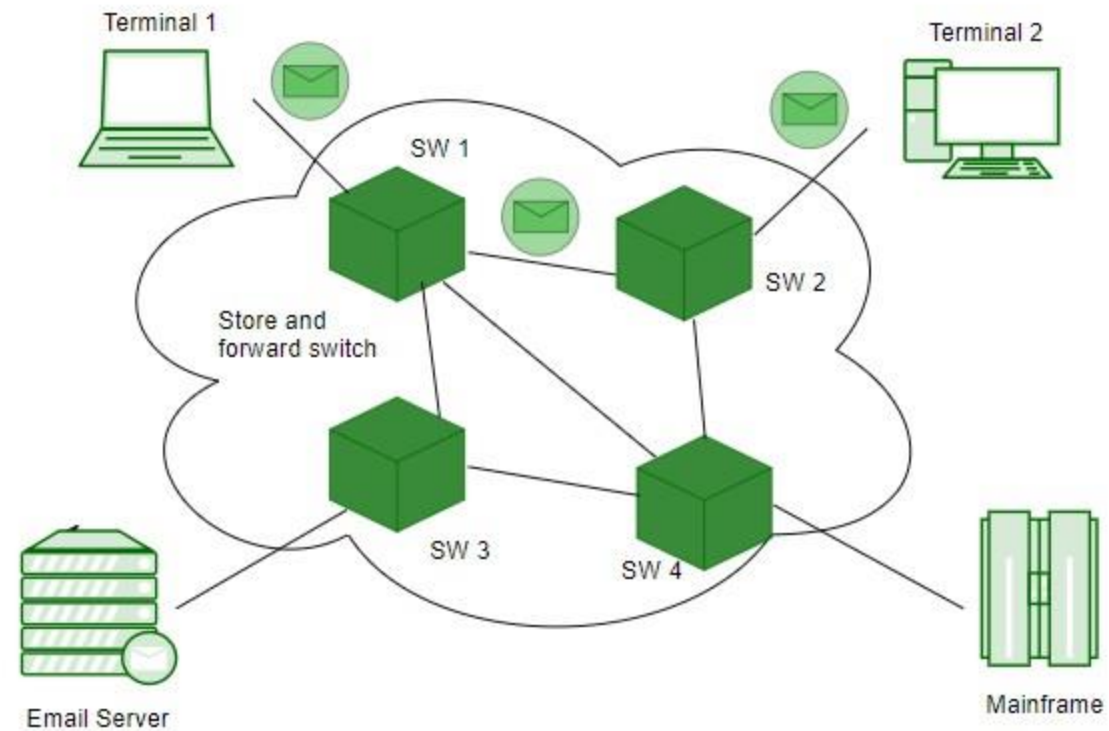
- ✓ Message Switching is a switching technique in which a message is transferred as a complete unit and routed through intermediate nodes at which it is stored and forwarded.
- ✓ In Message Switching technique, there is no establishment of a dedicated path between the sender and receiver.
- ✓ The destination address is appended to the message.
- ✓ Every node stores the entire message and then forward it to the next node. This type of network is known as store and forward network.
- ✓ They provide 2 distinct and important characteristics:

Message switching

Store and forward – The intermediate nodes have the responsibility of transferring the entire message to the next node.

Message delivery – This implies wrapping the entire information in a single message and transferring it from the source to the destination node.

Message switching



Message switching

In most electronic mail systems, the delivery process is based on message switching.

Advantages:

- ✓ It reduces network congestion.
- ✓ Better bandwidth usage.

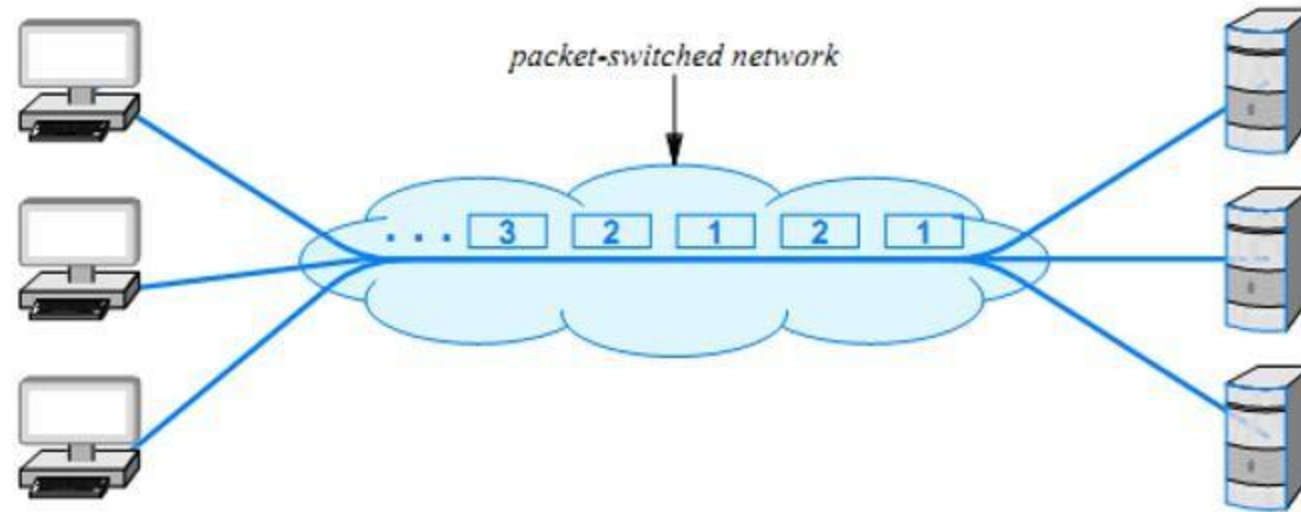
Disadvantages:

- ✓ Message switching cannot be used for real-time applications.
- ✓ Message has to be stored, for which every intermediate device in the network requires a large storage capacity.

3. Packet Switching

- ✓ The packet switching is a switching technique in which the message is sent in one go, but it is divided into smaller pieces, and they are sent individually.
- ✓ The message splits into smaller pieces known as packets and packets are given a unique number to identify their order at the receiving end.
- ✓ Every packet contains some information in its headers such as source address, destination address and sequence number.
- ✓ Packets will travel across the network, taking the shortest path as possible.
- ✓ All the packets are reassembled at the receiving end in correct order.
- ✓ If any packet is missing or corrupted, then the message will be sent to resend the message.
- ✓ If the correct order of the packets is reached, then the acknowledgment message will be sent.

3. Packet Switching



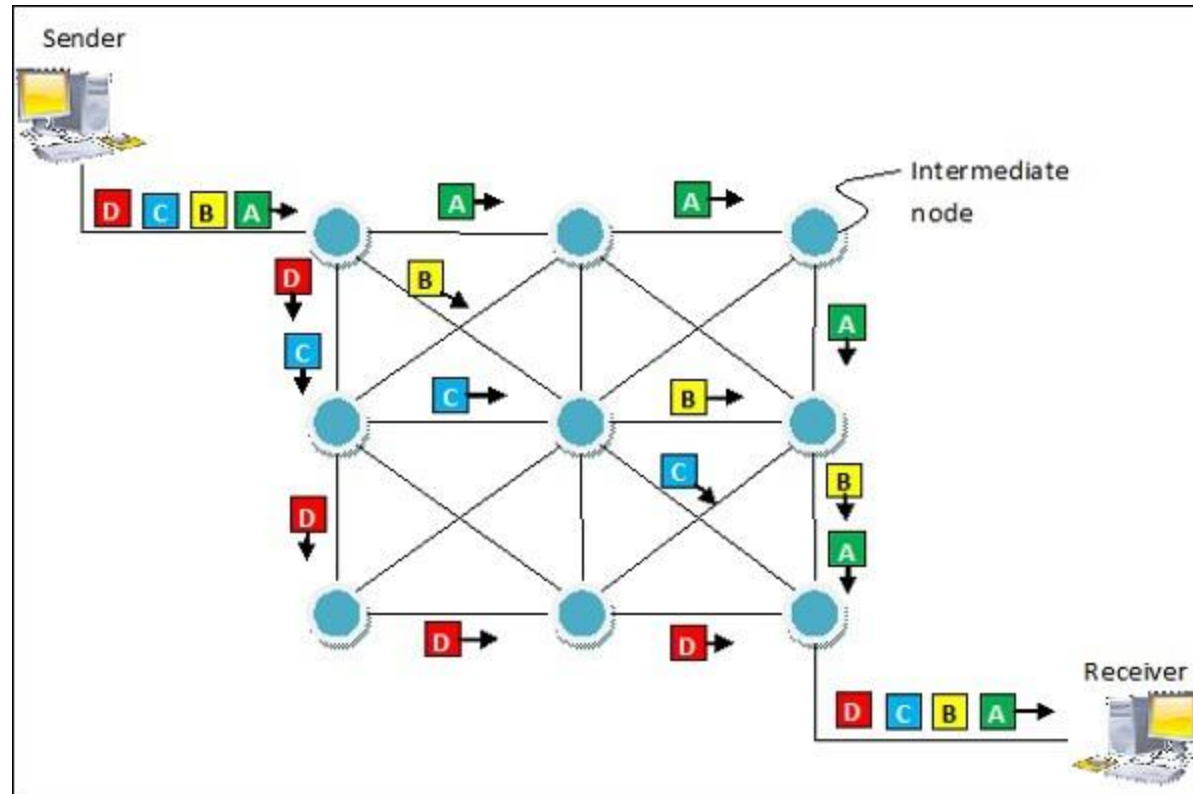
Approaches Of Packet Switching:

1. Datagram Packet Switching
2. Virtual Circuit Switching

1 . Datagram Packet Switching

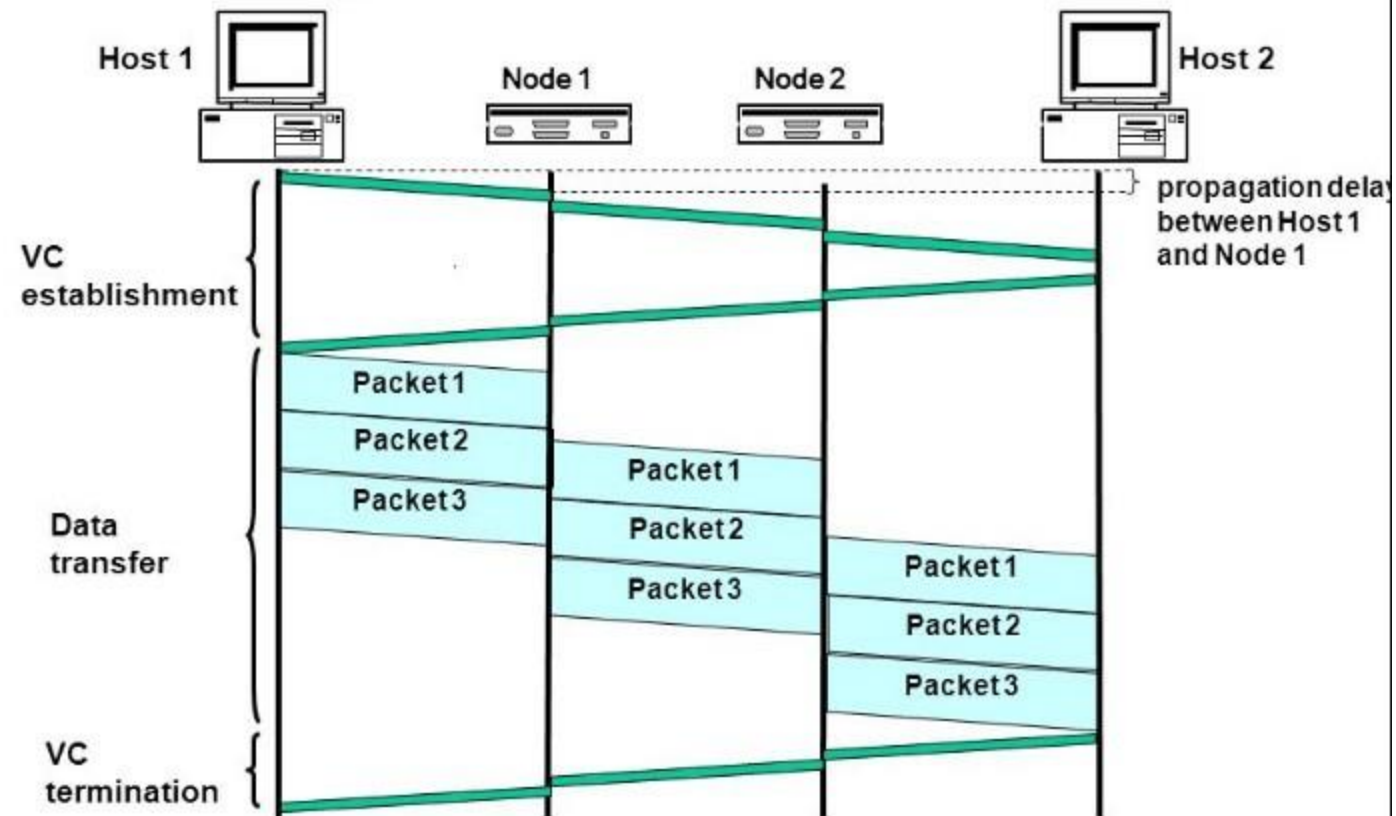
- ✓ It is a packet switching technology in which packet is known as a datagram, is considered as an independent entity. Each packet contains the information about the destination and switch uses this information to forward the packet to the correct destination.
- ✓ In Datagram Packet Switching technique, the path is not fixed.
- ✓ Intermediate nodes take the routing decisions to forward the packets.
- ✓ Datagram Packet Switching is also known as connectionless switching.

1. Datagram Packet Switching



2. Virtual Circuit Switching

- ✓ Virtual Circuit Switching is also known as connection-oriented switching.
- ✓ In the case of Virtual circuit switching, a preplanned route is established before the messages are sent.
- ✓ Call request and call accept packets are used to establish the connection between sender and receiver.
- ✓ In this case, the path is fixed for the duration of a logical connection.



Differences between Datagram approach and Virtual Circuit approach

<u>Datagram approach</u>	<u>Virtual Circuit approach</u>
• Intermediary node takes routing decisions to forward the packets.	• Intermediary node does not take any routing decision.
• Congestion cannot occur as all the packets travel in different directions.	• Congestion can occur when the node is busy, and it does not allow other packets to pass through.
• It is more flexible as all the packets are treated as an independent entity.	• It is not very flexible.

Advantages Of Packet Switching:

- ✓ Cost-effective: In packet switching technique, switching devices do not require massive secondary storage to store the packets, so cost is minimized to some extent.
- ✓ Reliable: It provides reliable communication.
- ✓ Efficient: It makes use of available bandwidth very efficiently.

Disadvantages Of Packet Switching:

- ✓ They are not suitable for applications that can't afford delays in communication like high quality voice calls.
- ✓ They involve high installation cost.
- ✓ They require complex protocols for delivery.
- ✓ If the network is overloaded or corrupted, then it requires retransmission of lost packets.

1 . Telephone Network:

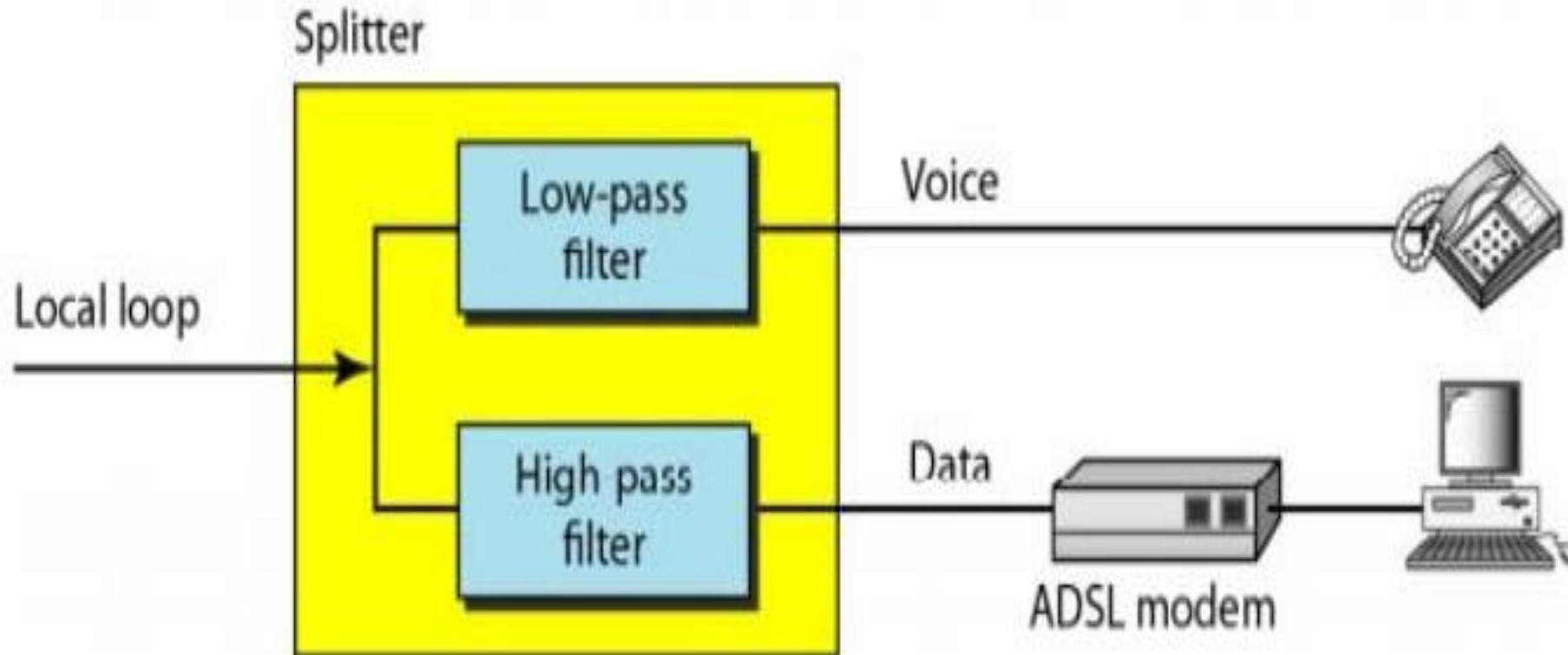
- ✓ Telephone Network is used to provide voice communication.
- ✓ Telephone Network uses Circuit Switching.
- ✓ Originally, the entire network was referred to as a plain old telephone system (POTS) which uses analog signals.
- ✓ The tasks of data transfer and signaling are separated in modern telephone network.

Major Components of Telephone Network:

There are three major components of the telephone network:

- a) Local loops
- b) Trunks
- c) Switching Offices

1. Telephone Network:



1. Telephone Network:

✓ They are the twisted pair cables that are used to connect a subscriber telephone to the nearest end office.

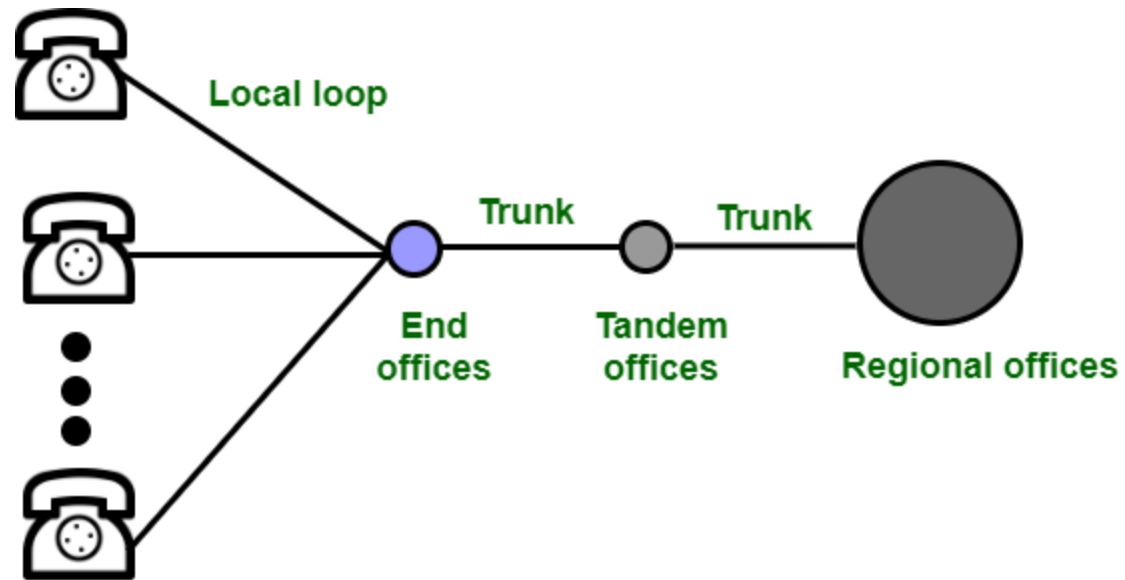
b) Trunks:

✓ They are transmission media used to handle the communication between offices.

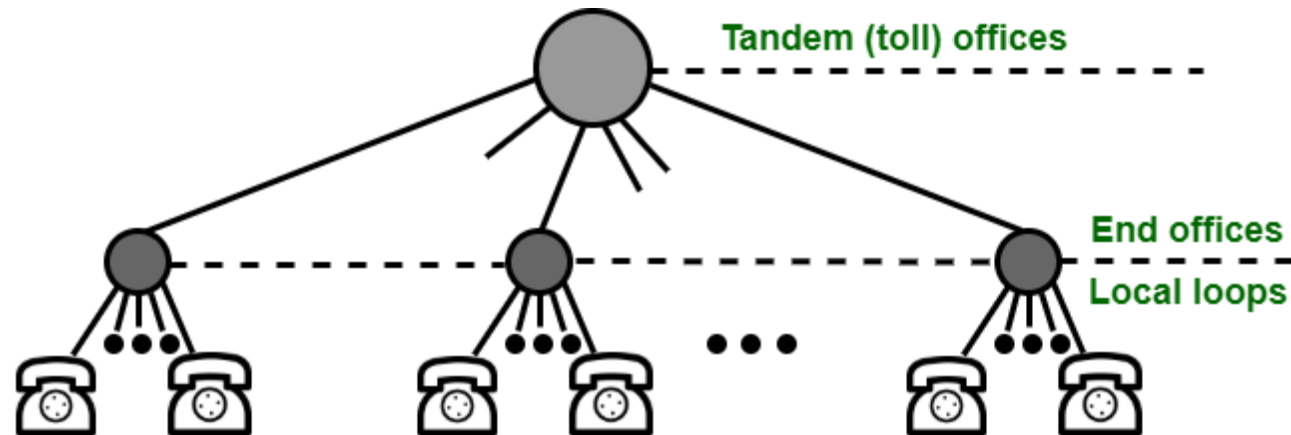
c) Switching offices:

✓ The telephone company uses switches which are able to connect various loops or trunks and allow a connection between different subscribers.

1. Telephone Network:



1. Telephone Network:



1. Telephone Network:

Advantages of the Telephone Network:

- ✓ It is a circuit-switched network.
- ✓ It is cheap in price because it is a widely spread network.

Disadvantages of the Telephone Network:

- ✓ It requires a large time for connection(in sec).
- ✓ It has a low transmission speed(64Kbps).

Applications of the Telephone Network:

- ✓ It helps to connect people.
- ✓ It is used by business organizations.

2. Cable Network:

- ✓ Cable TV network started its business only as a video service provider, but it's moved to the business of Internet access.
- ✓ It also refers to the system that distributes Television signals with the utilization of transmission media.

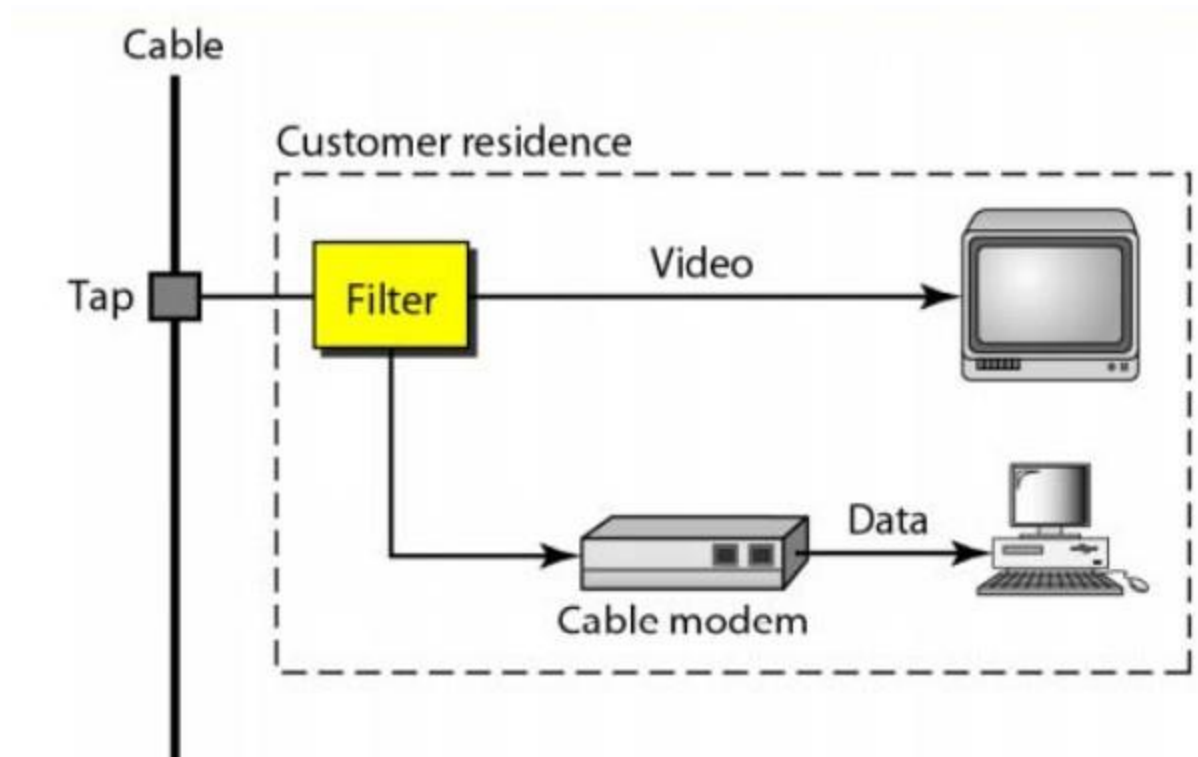
Types of Cable TV Networks are as follows:

- a) Traditional Cable Network
- b) Hybrid Fiber-Coaxial Network

a) Traditional Cable Network:

- ✓ This network began to distribute the broadcast video signals.
- ✓ The following is the schematic diagram of the traditional cable TV network.

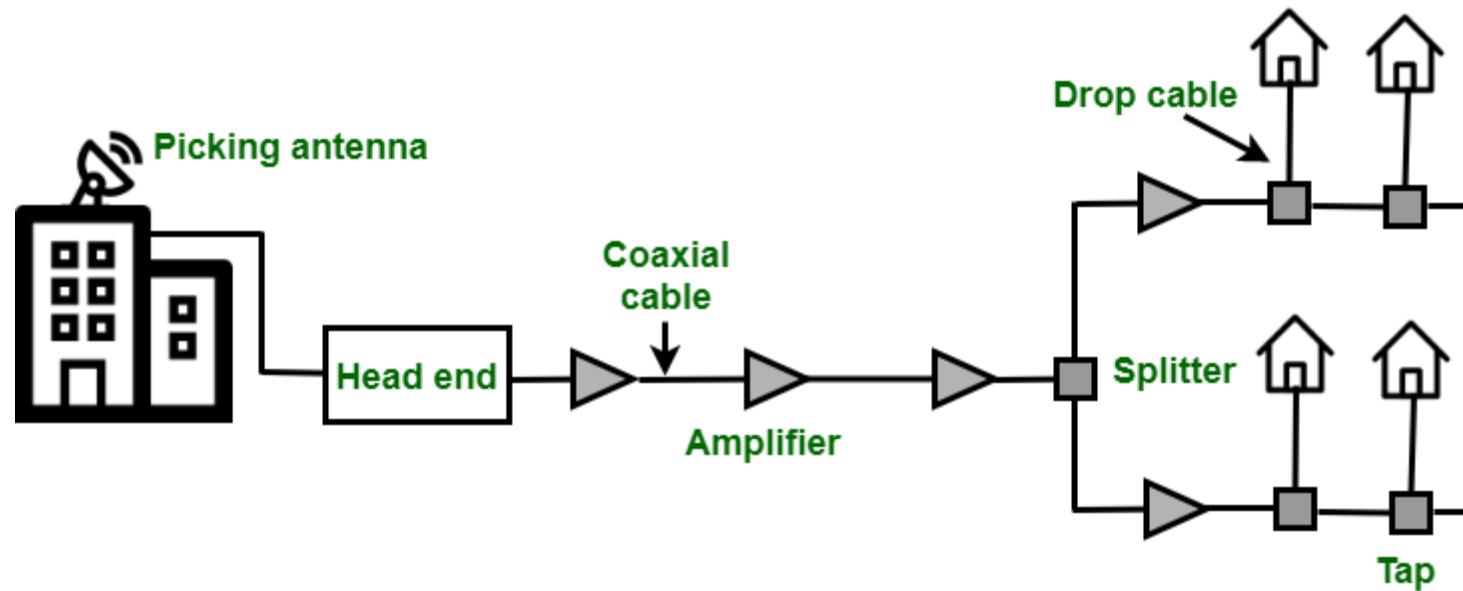
2. Cable Network:



2. Cable Network:

- ✓ Cable TV office is referred to as the head end.
- ✓ Amplifiers were installed through this network to regenerate the signals.
- ✓ Splitters were placed to split the signal and the tap and drop cables make the connections to the subscriber premises.

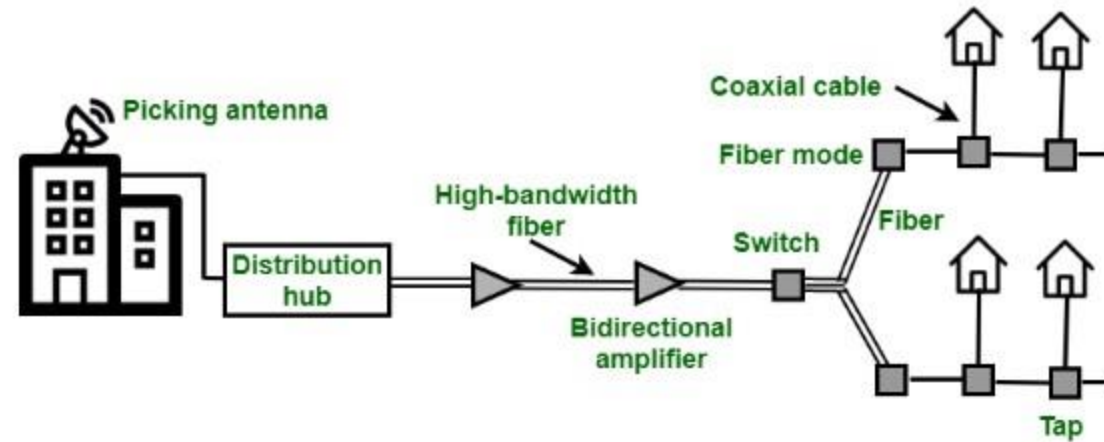
2. Cable Network:



b)Hybrid Fiber-Coaxial Network :

- ✓ Hybrid Fiber-Coaxial Network is the second generation of the cable network which is a combination of fiber-optic and coaxial cable.
- ✓ The fiber node split the analog signal so that the same signal is sent to each coaxial cable.
- ✓ The schematic diagram of the HFC network is as follows:

b)Hybrid Fiber-Coaxial Network :



b)Hybrid Fiber-Coaxial Network :

Advantages of Cable TV Network:

- ✓ Cable TV is stable in its service.
- ✓ It is fairly inexpensive.

Disadvantages of Cable TV Network:

- ✓ Due to the supply of single provider,it creates a monopoly(no competition).

3. Mobile Network:

✓ Mobile Data is the term used by many to describe the use of wireless data communications using micro waves to send and receive information.

Benefits of using mobile data communications:

- ✓ Communicate with people at remote locations.
- ✓ Communicate a message
- ✓ Keep in touch with the office anytime and from anywhere.

a) GSM

- ✓ GSM stands for Global System for Mobile communications.
- ✓ GSM is one of the most widely used digital wireless telephony system.
- ✓ GSM technology uses TDMA (Time Division Multiple Access).
- ✓ It also uses encryption to make the data more secure.
- ✓ The frequencies used by the international standard is 900 MHz to 1800 MHz.

b) CDMA:

- ✓ CDMA stands for Code Division Multiple Access.
- ✓ It employs spread spectrum technology.
- ✓ It allows several users to share a band of frequencies.
- ✓ The frequency range is 824 MHz to 894 MHz.

c) GPRS

- ✓ GPRS stands for General Packet Radio Services.
- ✓ It is a packet based wireless communication technology that charges users based on the volume of data they send.
- ✓ GPRS is used by second generation (2G) and third generation (3G) of mobile telephony.
- ✓ It pledges a speed of 56 kbps to 114 kbps, however the actual speed may vary depending on network load.

